The Impact of DeepSeek's Perceived Interactivity on Medical Students' Self-directed Learning Ability

Abstract

With the rapid advancement of artificial intelligence technology, DeepSeek, as a new-generation generative AI model, has demonstrated significant advantages in the field of medical education. Its robust interactive capabilities and localized deployment features make it particularly well-suited for medical education scenarios. This study aims to explore the mechanism and underlying pathways through which perceived interactivity influences medical students' self-directed learning ability. It also examines whether social influence indirectly affects self-directed learning ability via the mediating role of self-efficacy, and investigates whether trust moderates the relationship between social influence and behavioral intention. These findings reveal theoretical and practical implications for medical education contexts. This study employed SPSS 27.0 software for statistical data description, utilized Amos 27.0 software to validate the research model, and integrated Process 3.3.1 software to conduct moderation effect analysis. Building upon this foundation, an innovative research framework was constructed by synthesizing three major theoretical models. A random sampling method was used to collect 691 valid questionnaire responses from medical students. Structural equation modeling (SEM) and moderation effect analysis were then applied to test the research hypotheses. Perceived interactivity indirectly promotes willingness to use through performance expectancy (beta =

0.180, $p < 0.001$) and effort expectancy (beta = 0.428, $p < 0.001$), while social influence exerts the most significant direct effect on willingness to use (beta = 0.925, $p < 0.001$). Furthermore, self-efficacy played a crucial mediating role between intention to use and self-directed learning ability (beta = 0.575, $p < 0.001$), forming a psychological bridge from technology acceptance to capability enhancement. This study integrates the Unified Theory of Acceptance and Use of Technology (UTAUT), Social Cognitive Theory (SCT), and the Task-Technology Fit (TTF) model to construct a multidimensional mechanism framework examining how perceived interactivity of DeepSeek influences medical students' autonomous learning capabilities. This study not only validates the synergistic effects of social cognition and technological ease of use in the digital transformation of medical education but also provides theoretical support and practical pathways for the precise adaptation and optimization of DeepSeek within medical education settings. It offers significant implications for advancing the innovative development of medical education.